Recommended Assessment
Camera

Exploring Camera Sensors

1. What do you notice about the motion of the object?

   According to the OV5640 the camera sensor has a rolling shutter. Rolling shutter cameras will show visual artifacts because of the rolling affect used to update the image being captured.

2. What is the focal point of the camera? How did the estimated focal length vary from the physical length obtained from the lens spec sheet?

   Using the equation

   $$f_{mm} = f_{pxl}\, p_{xl} \tag{1}$$

   We can estimate the focal length in mm to be around.

   Using equation:

   $$FOV_{diag} = \arctan\left(\frac{\sqrt{w^2 + h^2}}{2f_{pxl}}\right) \tag{2}$$

   This gives an estimated 53 and a focal length of 1.11mm,

3. Change the environment (close a window, dim lights or add another light source). Were you still able to detect the color in the camera image?

   Using just RGB thresholding to detect a color will work only under certain conditions. To detect colors in different lighting conditions a more complex algorithm than RGB thresholding should be utilized. This is also very application dependent.

Real Life Scenarios

4. You are tasked to select a camera for incoming part inspection. Parts will be coming in from a conveyor belt and depending on the machine vision pipeline the parts will be rejected or not. What type of shutter and lens combination should the camera have?

   Since we're inspecting moving objects, we should use a global shutter camera to avoid visual artifacts as we're performing the incoming inspection. Depending on the inspection station a wide angle or standard lens could be used, however there will be significant distortion if a wide-angle camera is used.

5. A safety monitoring system uses a camera system to perform motion detection and detect intruders. You're asked to make sure the system works well in light and dark conditions. What are some considerations when deciding the camera system?

   Using a single RGB style sensor will only work during the day. During nighttime consider using a camera with a built-in infrared sensor to monitor a scene during low light conditions. This is an extension of the color section where we saw trying to track an object of a single color. When the lighting changes it becomes harder to track the object and a different type of camera sensor is needed.

Camera Cheat Sheet

What is it?	Converts light information into an image.
What does it measure directly?	The presence of red, green, blue values in the environment.
What can you measure indirectly?	Shapes, features in the environment. Type of inferred information depends on algorithm used to process image information.
Where are they used?	They are used in cellphones, autonomous vehicles, mobile robots, security systems, videography equipment, consumer electronics.
Pros	Passive sensor which uses low power to collect visual information from the environment.
Cons	Subject to environment lighting conditions, depending on the lens used distortion correction has to be performed before image-based algorithms are applied.
Notes/ Important things to remember	Cameras are useful assuming the lighting conditions remain constant. If there are large variations in environmental conditions, consider a fixture to ensure the camera image is constant or augmenting the camera with a different type of photoreceptive sensor.